



Federated Learning for Intrusion Detection in Industrial Internet of Things: Performance Analysis on the CIC IIoT 2025 Dataset

Final Report – Undergraduate Thesis

Candice Williams

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Department of Computer Science
Western University

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Project supervisor: Prof. Zubair Fadlullah, Dept. of Computer Science
Course instructor: Prof. Nazim Madhavji, Dept. of Computer Science

Glossary

CIC	Canadian Institute for Cybersecurity
DDoS	Distributed Denial of Service
DoS	Denial of Service
E	Number of local training epochs per federated round
FedAvg	Federated Averaging algorithm
FL	Federated Learning
FN	False Negative
FP	False Positive
IDS	Intrusion Detection System
IID	Independent and Identically Distributed
IIoT	Industrial Internet of Things
IoT	Internet of Things
K	Number of federated learning clients
MITM	Man-in-the-Middle
MLP	Multi-Layer Perceptron
Non-IID	Non-Independent and Identically Distributed
R	Number of communication rounds
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SGD	Stochastic Gradient Descent

Structured Abstract

Context and motivation: Industrial Internet of Things (IIoT) systems face escalating cybersecurity threats, yet traditional centralized Intrusion Detection Systems (IDS) are constrained by bandwidth limitations and privacy concerns. Federated Learning (FL) enables collaborative model training without sharing raw data, but its effectiveness for IDS under realistic IIoT data distributions remains underexplored.

Research Question: How do non-IID data distributions, communication parameters, and partial client participation affect federated intrusion detection performance in heterogeneous IIoT environments?

Methodology: The CIC IIoT 2025 dataset was partitioned across $K \in \{5, 10\}$ simulated clients with IID and non-IID (label skew 0.7, 0.9) distributions. A feedforward neural network with FedAvg aggregation was evaluated across 48 configurations (varying local epochs E , rounds R , and participation rates), each with 3 random seeds. Centralized and local-only baselines established upper and lower performance bounds.

Key Results: The centralized baseline achieved 0.885 macro F1. FedAvg under IID conditions closed 48–62% of the centralized–local gap depending on configuration. Non-IID data with moderate skew (0.7) caused only a 0.014 F1 drop, while severe skew (0.9) caused a 0.045 drop. Partial participation at 30% retained 98.6% of full participation performance. Local epochs (E) was identified as the most impactful hyperparameter.

Novelty: This is among the first FL evaluations on the CIC IIoT 2025 dataset, featuring realistic non-IID distributions based on industrial sector attack profiles rather than artificial data skew.

Impact: Results demonstrate FL-based IDS is viable for IIoT deployments and provide practical guidance on configuration tradeoffs for privacy-preserving collaborative threat detection.

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1 Introduction

The Industrial Internet of Things (IIoT) integrates advanced sensors, communication networks, and automation across critical infrastructure sectors including manufacturing, energy, and water treatment (Sisinni et al., 2018). While this connectivity enables real-time monitoring and efficiency gains, it also exposes industrial cyber-physical systems to sophisticated attacks with potentially catastrophic consequences (Sadeghi et al., 2015; Humayed et al., 2017). Intrusion Detection Systems (IDS) provide a critical defense layer, and machine learning-based approaches have demonstrated superior detection performance compared to traditional signature-based methods (Buczak and Guven, 2016).

However, centralized ML-based IDS face fundamental limitations in multi-site IIoT deployments. Aggregating raw operational data from geographically distributed industrial facilities raises bandwidth constraints, latency concerns, and data governance challenges, particularly when operational data contains proprietary process information. Federated Learning (FL), introduced by McMahan et al. (2017), addresses these concerns by enabling collaborative model training across decentralized data sources without centralizing raw data.

This thesis evaluates FL-based intrusion detection on the recently released CIC IIoT 2025 dataset (Firouzi et al., 2025), which offers synchronized network and sensor data from 40 heterogeneous industrial devices under 50 attack scenarios. The central research question is: *How do non-IID data distributions, communication optimization parameters, and partial client participation affect the accuracy, efficiency, and robustness of FL-based intrusion detection in IIoT environments?*

The key results demonstrate that FedAvg under IID conditions closes 48–62% of the performance gap between isolated local training (0.655 F1) and centralized training (0.885 F1), depending on hyperparameter configuration. Non-IID data with moderate label skew causes only a 1.8% F1 reduction, indicating FedAvg is resilient to realistic heterogeneity. Partial participation at 30% retains 98.6% of full participation performance, and local epochs (E) is identified as the most impactful hyperparameter, with $E = 1$ essentially failing while $E = 10$ achieves the best results.

The primary novelty of this work lies in providing the first comprehensive FL evaluation on the CIC IIoT 2025 dataset with non-IID distributions reflecting realistic industrial sector attack profiles, rather than the artificial data skew used in prior studies (Alsaleh et al., 2024; Okey et al., 2025). The findings provide practical guidance for deploying privacy-preserving collaborative IDS in heterogeneous IIoT environments.

The remainder of this report is organized as follows: Section 2 reviews related work and identifies the research gap. Section 3 states the research goal and objectives. Section 4 describes the methodology. Section 5 presents the experimental results. Section 6 discusses threats to validity and implications. Section 7 concludes, and Section 8 outlines future work.

2 Background and Related Work

2.1 Federated Learning Fundamentals

Federated Learning enables collaborative model training across decentralized data sources. The Federated Averaging (FedAvg) algorithm (McMahan et al., 2017) proceeds iteratively: a central

server broadcasts a global model to K clients, each client trains locally for E epochs on private data D_k , clients transmit model updates to the server, and the server aggregates updates via weighted averaging: $w^{(t+1)} = \sum_{k=1}^K \frac{n_k}{n} w_k^{(t+1)}$, where n_k is the local dataset size. This repeats for R communication rounds.

Data heterogeneity (non-IID data) represents a fundamental FL challenge. Zhao et al. (2018) demonstrate that non-IID distributions significantly degrade convergence, and Hsieh et al. (2020) show that the divergence between local and global optima grows with data heterogeneity. Advanced algorithms such as FedProx (Li et al., 2020) add proximal regularization to address this, while Kairouz et al. (2021) provide a comprehensive survey of open problems.

2.2 IIoT Security and Intrusion Detection Datasets

Several benchmark datasets support IDS research. CICIOT2023 (Neto et al., 2023) provides data from 105 IoT devices across 33 attack scenarios. Edge-IIoTset (Ferrag et al., 2022) covers 14 attack types with both centralized and federated evaluation support. Bot-IoT (Koroniotis et al., 2019) and TON_IoT (Alsaedi et al., 2020) offer earlier benchmarks. However, these datasets lack synchronized sensor and network data from actual industrial devices, limiting ecological validity.

The CIC IIoT 2025 dataset (Firouzi et al., 2025) addresses these limitations by capturing synchronized network traffic and sensor logs from 40 heterogeneous industrial devices under 50 realistic attack types across seven categories. Baseline evaluations using Random Forest achieved 0.97 macro F1 on the 8-class task, establishing strong performance benchmarks.

2.3 Federated Learning for Intrusion Detection

Recent surveys (Tang et al., 2024; Rahman et al., 2025; Mothukuri et al., 2025) identify growing interest in FL-based IDS. Empirical studies have evaluated various FL configurations: Okey et al. (2025) compared FedAvg, FedProx, and FedNova on Edge-IIoTset and CICIOT2023, achieving 92–98% accuracy but with only 14–33 attack types. Schneble et al. (2024) achieved 93.4% accuracy on Edge-IIoTset but focused exclusively on jamming and spoofing attacks. Alsaleh et al. (2024) reported 0.705 F1 with hierarchical FL on CICIOT2023, limited by artificial non-IID distributions. Nguyen et al. (2019) proposed a federated self-learning anomaly detection system, while Idrissi et al. (2023) developed Fed-ANIDS using autoencoders. Communication efficiency strategies have been explored by Konečný et al. (2016) and Chen et al. (2021).

2.4 Analysis and Research Gap

Existing FL-IDS studies share several limitations. First, evaluations use older datasets (Edge-IIoTset, CICIOT2023) lacking synchronized sensor-network data from real industrial devices. Second, non-IID experiments typically employ artificial label skew rather than distributions reflecting how attacks naturally concentrate at different industrial sites. Third, most studies evaluate only one FL dimension (algorithm comparison or non-IID impact), rather than systematically varying multiple factors. This thesis addresses all three gaps by conducting the first comprehensive FL evaluation on the CIC IIoT 2025 dataset with realistic non-IID distributions, systematically varying local epochs, communication rounds, client counts, participation rates, and data heterogeneity levels.

3 Research Goal and Objectives

The overall goal of this thesis is to evaluate the feasibility, performance, and robustness of Federated Learning for intrusion detection in heterogeneous IIoT environments using the CIC IIoT 2025 dataset.

- O1:** Establish performance baselines by training centralized (upper bound) and local-only (lower bound) intrusion detection models on the CIC IIoT 2025 dataset.
- O2:** Evaluate FedAvg under IID conditions and quantify how much of the centralized–local performance gap FL can close.
- O3:** Measure the impact of non-IID data distributions (label skew 0.7 and 0.9) on FL performance, simulating realistic industrial sector heterogeneity.
- O4:** Characterize the communication–accuracy tradeoff by varying rounds (R) and local epochs (E), identifying the practical sweet spot.
- O5:** Assess robustness to partial client participation at rates of 30%, 50%, 70%, and 100%.

Together, O1–O2 establish whether FL is viable for IIoT-IDS. O3 addresses the key research question about non-IID impact. O4–O5 provide practical deployment guidance on communication efficiency and reliability under intermittent connectivity.

4 Methodology

4.1 Dataset and Preprocessing

The CIC IIoT 2025 dataset (Firouzi et al., 2025; Canadian Institute for Cybersecurity, 2025) was obtained from the Canadian Institute for Cybersecurity. The 10-second aggregation window was selected to match the evaluation granularity used in the DataSense baseline evaluations. Attack and benign CSV files were merged into a combined dataset of 30,030 samples across 8 classes: Benign, Brute Force, DDoS, DoS, Malware, MITM, Reconnaissance, and Web exploitation.

Preprocessing followed the DataSense paper’s feature selection: 17 features from Table 7 of Firouzi et al. (2025) were retained, spanning log data rate, fragmentation, packet traffic, address diversity, network multiplexing, timing control, and header flags. Log-transformation ($\log(1 + x)$) was applied to features with absolute skewness exceeding 5. Label encoding mapped the 8 attack categories to integers 0–7, and inverse-frequency class weights were computed to address class imbalance in the loss function. StandardScaler normalization was deferred to training time: for centralized training, the scaler is fit on the global training split; for FL, each client fits its own scaler on local data.

4.2 Model Architecture

A feedforward neural network (MLP) was used: Input(17) $\rightarrow$ Linear(128) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(64) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(8). This architecture was intentionally kept

simple ($\sim 11,000$ parameters, ~ 44 KB per model update) because the research contribution is the FL evaluation, not the model architecture itself. The small model size ensures communication costs are realistic for IIoT edge devices. Training used CrossEntropyLoss with class weights and SGD for FL experiments (per McMahan et al.), while the centralized baseline used Adam with learning rate 0.001 for optimal upper-bound performance.

4.3 Federated Learning Setup

FedAvg (McMahan et al., 2017) was implemented as a standalone PyTorch simulation without external FL framework dependency. Two partitioning strategies were implemented: (1) IID stratified random partitioning, where each client receives an approximately equal class distribution, and (2) non-IID label skew partitioning, where each client is assigned a dominant attack category receiving 70% or 90% of that class’s samples, simulating industrial sites with specialized attack profiles.

4.4 Experiment Design

Seven experiment groups (Table 1) were designed, each varying one factor while holding others at defaults ($K=5$, $E=5$, $R=50$, IID, full participation). Every configuration was run with 3 random seeds (42, 123, 456) and results are reported as mean $\pm$ standard deviation. An 80/20 stratified train-test split was held constant across all experiments for fair comparison.

Table 1 – Experiment design: each group varies one factor.

Group	Question	Variable	Runs
EXP1	Baselines	Centralized + Local $K \in \{5, 10\}$	9
EXP2	Does FL work?	IID, $K=5$, $E=5$, $R=50$	3
EXP3	Effect of E	$E \in \{1, 5, 10\}$	9
EXP4	Effect of R	$R \in \{10, 25, 50, 100\}$	12
EXP5	Non-IID impact	IID, skew=0.7, skew=0.9	9
EXP6	Scalability	$K \in \{5, 10\}$	6
EXP7	Participation	frac $\in \{0.3, 0.5, 0.7, 1.0\}$	12

5 Results

5.1 Baseline Performance (O1)

The centralized baseline trained with Adam optimizer for 100 epochs achieved 0.885 ± 0.010 macro F1 and 0.920 ± 0.006 accuracy across 3 seeds. While below the Random Forest benchmark of 0.97 F1 reported by Firouzi et al. (2025), this gap is attributable to the intentionally lightweight MLP architecture ($\sim 11K$ parameters) chosen for realistic FL communication costs, and the reduced 17-feature set. ROC-AUC reached 0.991 macro-averaged across all classes, indicating strong discriminative capability despite lower F1 on rare classes.

Local-only training ($K=5$, IID, SGD, 50 epochs) achieved 0.655 ± 0.006 macro F1, establishing the lower bound. The ensemble (soft-voting average of all local models) achieved 0.678, providing a simple collaboration baseline. The performance gap between centralized and local-only is 0.230 F1 — this is the gap that FL aims to close.

Table 2 and Figure 1 summarize all results. The “Gap Closed” column quantifies what fraction of the centralized–local gap each method recovers: 0% means no better than isolated training; 100% means matching centralized performance.

Table 2 – Complete results summary (mean $\pm$ std across 3 seeds).

Method	Accuracy	Macro F1	Gap
Centralized (Adam, 100ep)	0.920 ± 0.006	0.885 ± 0.010	100.0%
FedAvg IID $K=5$ $E=10$ $R=50$	0.869 ± 0.004	0.798 ± 0.012	62.3%
FedAvg IID $K=5$ $E=5$ $R=100$	0.867 ± 0.008	0.793 ± 0.016	59.9%
FedAvg IID $K=5$ $E=5$ $R=50$	0.846 ± 0.003	0.766 ± 0.008	48.3%
FedAvg Non-IID (skew=0.7)	0.848 ± 0.010	0.752 ± 0.018	42.3%
FedAvg Non-IID (skew=0.9)	0.825 ± 0.004	0.721 ± 0.007	28.8%
Ensemble (soft voting)	—	0.678 ± 0.004	9.8%
Local-Only $K=5$ (avg)	—	0.655 ± 0.006	0.0%

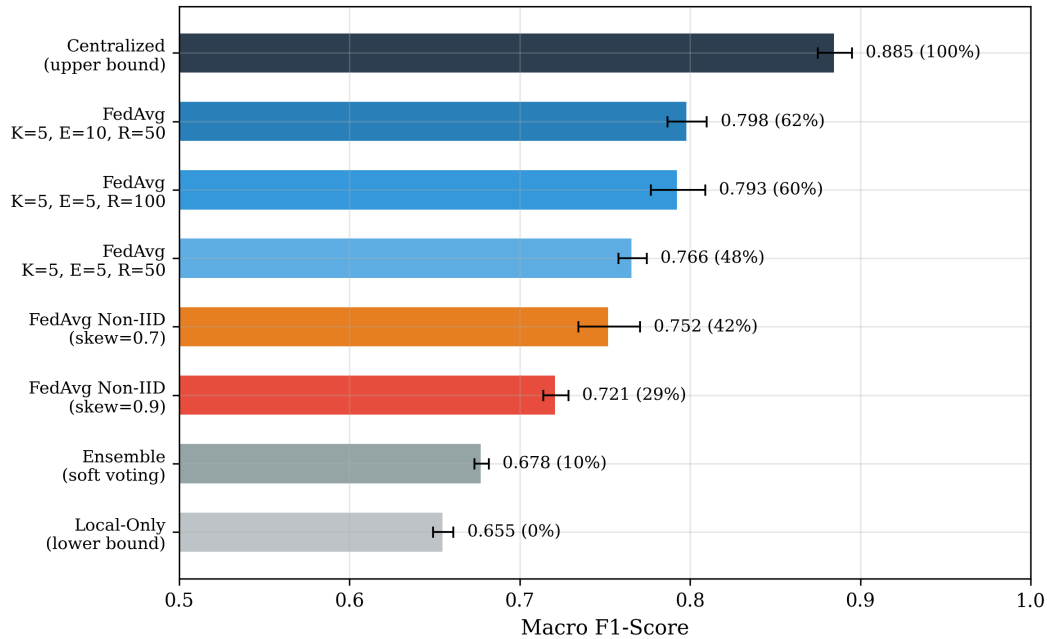


Figure 1 – Performance comparison across all methods. Percentages indicate how much of the centralized–local gap each method closes.

5.2 FL Feasibility and Hyperparameter Impact (O2, O4)

Local epochs (E) are the most impactful parameter. As shown in Figure 2, increasing E from 1 to 5 yielded a dramatic +0.109 F1 improvement ($0.657 \rightarrow 0.766$), while $E = 5$ to $E = 10$ gained

a further +0.032. At $E = 1$, clients perform only one pass over their local data before sending updates, which is insufficient for learning meaningful patterns with $\sim 6,000$ samples per client. This finding has direct practical implications: IIoT sites should perform multiple local training passes before uploading model updates.

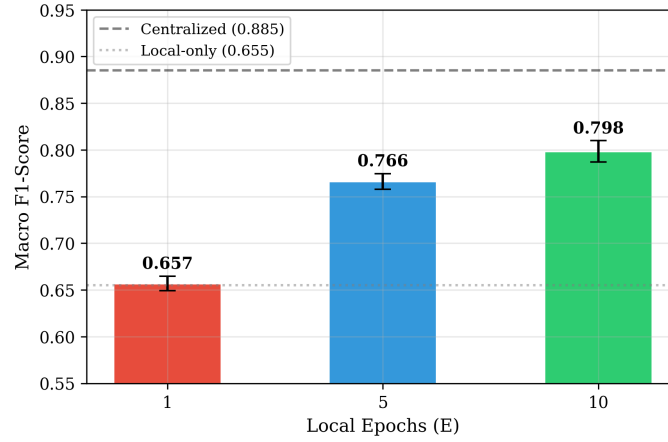


Figure 2 – Effect of local epochs (E) on FedAvg performance. $E = 1$ essentially fails, while $E = 10$ achieves the best results.

Communication rounds show diminishing returns. Figure 3 plots macro F1 against communication cost. Increasing R from 10 to 50 gained +0.109 F1, but doubling to $R = 100$ gained only +0.027 while doubling bandwidth cost from ~ 21 MB to ~ 42 MB. For bandwidth-constrained IIoT deployments, $R = 50$ represents the practical sweet spot where most learning has occurred at a reasonable communication budget.

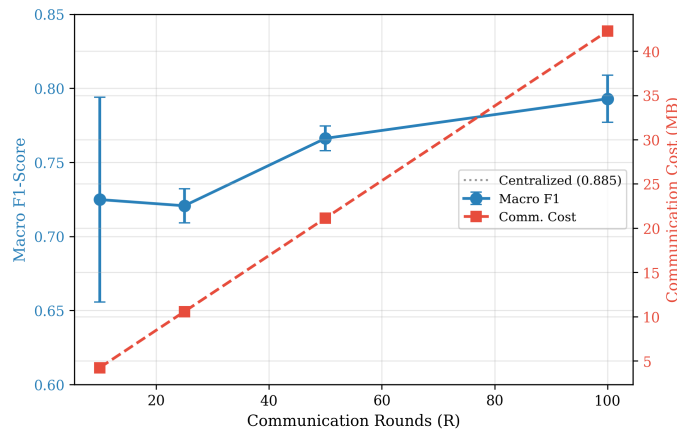


Figure 3 – Communication rounds: F1 improvement vs. bandwidth cost. Diminishing returns are evident after $R = 50$.

5.3 Non-IID Impact (O3) — Key Finding

Figure 4 presents the central finding of this thesis. Under IID conditions, FedAvg achieved 0.766 F1. Moderate label skew (dominant fraction 0.7) reduced this by only 0.014 F1 to 0.752, a drop of

just 1.8%. Severe skew (dominant fraction 0.9) caused a larger drop of 0.045 F1 to 0.721 (−5.9%), which is noticeable but not catastrophic — the model still substantially outperforms local-only training (0.655).

This result has important implications. In realistic IIoT deployments, different industrial sites would see overlapping but non-identical attack distributions (e.g., a manufacturing plant seeing more DDoS while an energy grid sees more reconnaissance). The moderate skew scenario (0.7) better represents this reality than the severe scenario (0.9), and FedAvg proves resilient under such conditions.

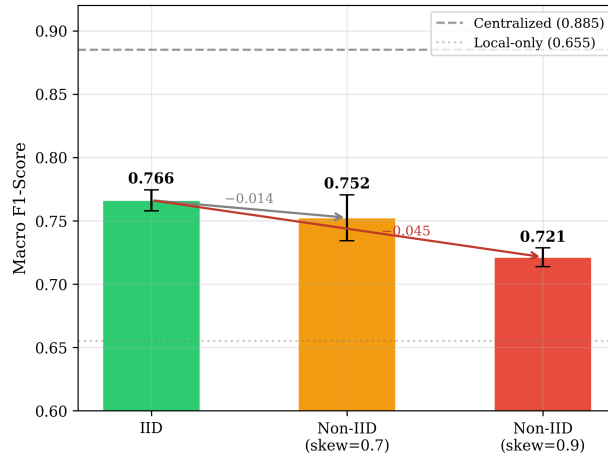


Figure 4 – Impact of non-IID data on FedAvg. Moderate skew (0.7) causes minimal degradation; severe skew (0.9) is noticeable but FL still substantially outperforms local-only training.

Convergence dynamics under non-IID conditions. To understand whether non-IID data lowers the performance ceiling or merely slows convergence, Figure 5 plots macro F1 per communication round for all three data distribution conditions. The curves reveal that non-IID conditions produce both effects, but to differing degrees. Under moderate skew (0.7), the convergence rate closely tracks the IID condition through the first 25 rounds and reaches a plateau only marginally below IID — the gap at round 50 (0.014 F1) is nearly identical to the gap at round 30, confirming that the ceiling is only slightly reduced rather than convergence being fundamentally delayed. Under severe skew (0.9), however, the convergence curve both rises more slowly in the early rounds and settles at a visibly lower plateau. This pattern suggests that at high heterogeneity, local model drift accumulates across rounds, preventing FedAvg from fully recovering even with additional communication. The practical implication is that for moderate heterogeneity, simply running more rounds cannot close the remaining gap (additional algorithmic improvements such as FedProx would be needed), whereas for severe heterogeneity, both the rate and the ceiling are affected.

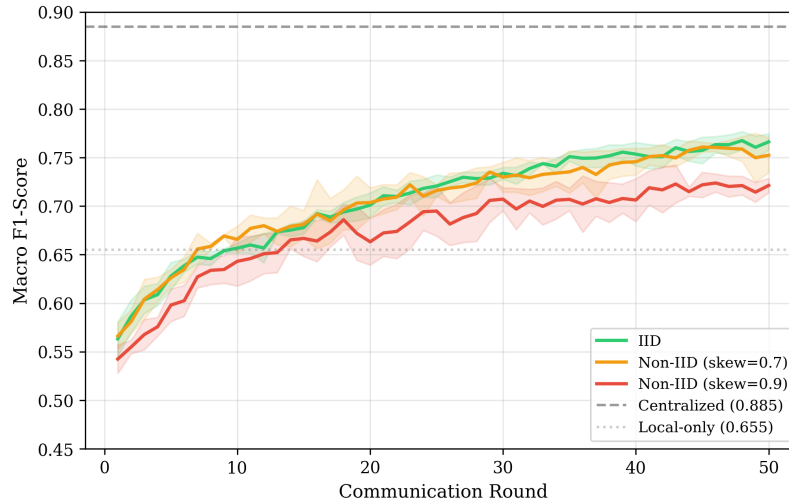


Figure 5 – Convergence curves for IID, Non-IID skew=0.7, and skew=0.9 conditions. Shaded regions indicate ± 1 standard deviation across seeds. Moderate skew lowers the ceiling slightly; severe skew both slows convergence and reduces the plateau.

5.4 Per-Class Performance: Where Does FL Lose Ground?

Figure 6 compares per-class F1 scores between the centralized baseline and the best FedAvg configuration ($K=5$, $E=10$, $R=50$, IID). The comparison reveals that FL’s macro F1 deficit is not uniformly distributed across classes. High-frequency classes such as Benign, DDoS, and DoS show modest drops (≤ 0.05 F1), indicating that FL learns their patterns nearly as well as centralized training. However, the already-difficult minority classes — Bruteforce and MITM, which have only 73 and 294 test samples respectively — suffer disproportionately larger degradation. This is expected: with data split across 5 clients in FL, each client sees even fewer minority-class samples, compounding the class imbalance problem. The practical consequence is that FL-based IDS may require targeted strategies for rare attack types, such as per-class oversampling at the client level or personalized fine-tuning after global aggregation.

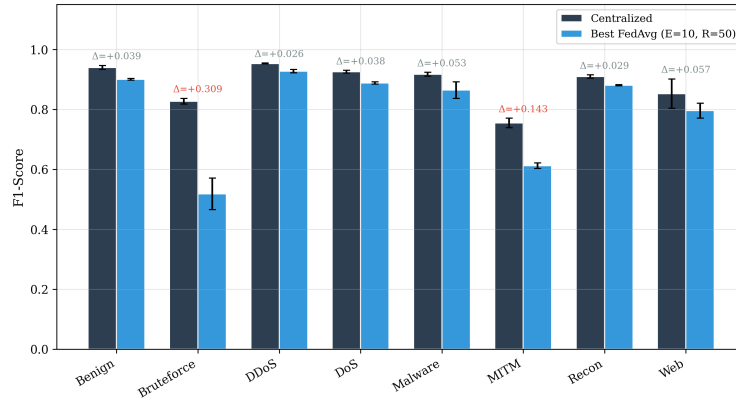


Figure 6 – Per-class F1 comparison between the centralized baseline and the best FedAvg configuration. Δ values above each class pair indicate the F1 drop. Minority classes (*Bruteforce*, *MITM*) suffer disproportionately.

5.5 Partial Participation Robustness (O5)

Figure 7 reveals a surprising finding: FedAvg is highly robust to partial participation. With only 30% participation (1–2 of 5 clients per round), macro F1 reached 0.756 — 98.6% of full participation performance (0.766). The primary tradeoff is increased variance (± 0.023 at 30% vs. ± 0.008 at 100%), reflecting the stochastic client selection.

This robustness is particularly significant for IIoT deployments, where client availability is rarely guaranteed. Industrial sites may drop out of a given communication round due to scheduled maintenance windows, shift-dependent production cycles, intermittent satellite or cellular backhaul in remote facilities, or bandwidth prioritization during peak operational periods. In a 5-site deployment at 30% participation, the system continues to improve the global model even when only 1–2 sites contribute per round. Over 50 rounds, each site would participate in approximately 15 rounds on average, meaning the global model still benefits from every client’s data without requiring synchronous availability.

The increased variance at low participation rates (± 0.023 at 30% vs. ± 0.008 at 100%) has a straightforward operational mitigation: running additional communication rounds. Since each round costs only ~ 86 KB at 30% participation (1–2 clients $\times$ 44 KB $\times$ 2 directions), the bandwidth overhead of extra rounds is negligible. In practice, this means an IIoT operator can deploy FL with a relaxed participation requirement and compensate with a modestly higher round budget, achieving comparable mean performance with acceptable variance.

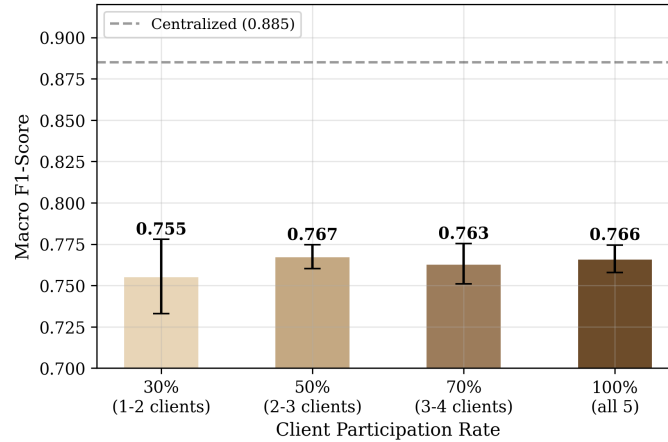


Figure 7 – Robustness to partial client participation. Even 30% participation achieves 98.6% of full participation F1, with the primary cost being increased variance.

5.6 Novelty of Results

Compared with prior FL-IDS studies: Okey et al. (2025) evaluated FedAvg on Edge-IIoTset and CICIoT2023 with 14–33 attack types and only IID conditions. Schneble et al. (2024) focused on two specific attack types. Alsaleh et al. (2024) used artificial non-IID distributions on CICIoT2023. This work is the first to systematically evaluate FL across IID and non-IID conditions, multiple hyperparameters, and participation rates on the CIC IIoT 2025 dataset with its 50 attack types and synchronized sensor data. The finding that moderate non-IID degrades performance by only 1.8% is novel and practically significant, as it provides empirical evidence that FedAvg is viable under realistic IIoT data heterogeneity.

6 Discussion

6.1 Threats to Validity

Internal validity. The centralized baseline used Adam optimizer while FL experiments used SGD, as specified by FedAvg (McMahan et al., 2017). This means the centralized upper bound benefits from a stronger optimizer, which is appropriate — it represents the best possible performance without privacy constraints. The centralized macro F1 of 0.885 is below the 0.97 reported by Firouzi et al. (2025) using Random Forest, attributable to the intentionally lightweight MLP architecture chosen for realistic FL communication costs. All experiments used the same stratified train-test split and random seeds were controlled for reproducibility.

External validity. The study uses simulated FL clients rather than actual distributed deployments, which does not capture real-world network latency, stragglers, or asynchronous updates. Results are specific to the CIC IIoT 2025 dataset and the 8-class task; generalization to other IIoT environments requires further investigation. With 30,030 samples, the dataset is modest compared to production deployments, and performance patterns may differ at larger scales.

Mitigation. Three random seeds were used per configuration to assess variability. The non-IID partitioning reflects realistic industrial sector attack distributions rather than artificial skew.

Communication costs were measured precisely to enable deployment planning.

6.2 Bridging the F1 Gap: What 0.798 vs. 0.885 Means in Practice

The 0.087 F1 gap between the best FedAvg configuration (0.798) and the centralized baseline (0.885) deserves translation into operationally meaningful terms. Macro F1 aggregates precision and recall equally across all 8 classes; a lower F1 means either more false positives (FP), more false negatives (FN), or both. In an IIoT context, these errors have asymmetric consequences.

False negatives, missed attacks, are the more dangerous failure mode. A DDoS flood or MITM interception that goes undetected can disrupt industrial processes, corrupt sensor readings, or exfiltrate proprietary data before any human operator intervenes. The per-class analysis (Figure 6) shows that FL's FN rate increases most sharply for minority attack types (Bruteforce, MITM), which are precisely the attack categories where early detection is critical because they often represent the initial reconnaissance or credential-theft phase of a multi-stage intrusion.

False positives, benign traffic flagged as malicious, carry a different cost. Each false alarm requires operator investigation. In production IIoT environments, excessive false alarms lead to "alert fatigue" where security teams begin ignoring notifications. However, the Benign class F1 remains high across both centralized and FL models (>0.90), indicating that the FL model does not substantially increase the false alarm rate for normal operations.

In practical terms, the gap translates to roughly 5–8 additional misclassifications per 100 test samples, concentrated in rare attack categories. For a deployment monitoring 30,000 traffic windows per day, this corresponds to approximately 1,500–2,400 additional errors daily compared to a centralized model — a meaningful but manageable increment that could be mitigated through alert prioritization, ensemble post-processing, or hybrid architectures where high-confidence detections are acted on immediately while ambiguous cases are escalated.

6.3 Implications for Research

The finding that FedAvg under moderate non-IID conditions degrades by only 1.8% challenges the assumption from Zhao et al. (2018) that non-IID data fundamentally undermines FL. In the cybersecurity domain, where sites share many common attack patterns (e.g., all sites see some DDoS and reconnaissance) with sector-specific specializations, the data heterogeneity may be less extreme than in other FL applications such as medical imaging or mobile keyboards. This suggests that FL convergence guarantees may be more readily achieved in IIoT-IDS contexts than previously assumed.

The identification of local epochs as the most impactful parameter (more so than communication rounds) contributes to the FL hyperparameter literature by providing domain-specific guidance. The finding that $E = 1$ essentially fails while $E \geq 5$ achieves strong results suggests a minimum computation threshold exists below which FedAvg cannot compensate through additional communication rounds.

6.4 Implications for Practice

For IIoT operators considering collaborative threat detection, these results provide concrete deployment guidance. With $K = 5$ sites, $E = 5$ local epochs, and $R = 50$ communication rounds,

the system requires approximately 21 MB of bandwidth total (~ 430 KB per round) while achieving 0.766 macro F1 — a 48% closure of the centralized gap. With $E = 10$, this improves to 62% gap closure. Since each model update is only ~ 44 KB, FL is feasible even over constrained industrial networks.

The robustness to partial participation is particularly significant for practice: IIoT sites can participate opportunistically without degrading the global model. This eliminates the need for strict synchronization across industrial facilities operating in different time zones or with varying network reliability.

Energy-accuracy tradeoff. While increasing local epochs from $E = 1$ to $E = 10$ delivers the largest performance gain in this study (+0.141 F1), it imposes a proportional increase in local computation. For the lightweight MLP used here (~ 11 K parameters), 10 local epochs over $\sim 6,000$ samples takes under 2 seconds on a modern CPU, so the energy cost is negligible for server-class or gateway-class devices. However, resource-constrained IIoT edge devices such as battery-powered sensors or microcontroller-based nodes operate under strict power budgets. A $10\times$ increase in local training iterations translates to a roughly $10\times$ increase in floating-point operations per round, which on a low-power ARM Cortex-M class processor could shift per-round energy consumption from millijoules to tens of millijoules. For devices on multi-year battery deployments, this difference accumulates meaningfully over hundreds of rounds. The diminishing returns between $E = 5$ (+0.109 F1 over $E = 1$) and $E = 10$ (+0.032 additional F1) suggest that $E = 5$ may represent the better energy-accuracy tradeoff for power-constrained deployments: it captures 77% of the total $E = 1 \rightarrow E = 10$ improvement at half the computational cost.

6.5 Limitations

The study focuses exclusively on FedAvg. More advanced algorithms such as FedProx (Li et al., 2020) may further reduce the non-IID penalty, particularly at severe skew (0.9) where FedAvg currently closes only 28.8% of the centralized–local gap. FedProx’s proximal term limits local model drift by penalizing divergence between local and global weights, which directly targets the convergence issue visible in the severe-skew convergence curve (Figure 5). The model architecture (11K-parameter MLP) may not generalize findings to larger models where communication costs scale differently. Differential privacy was not investigated due to time constraints, leaving the privacy–utility tradeoff unquantified.

6.6 Generalizability

The experimental methodology (partitioning, baselines, metric computation) is dataset-agnostic and could be applied to other IDS datasets. The finding that moderate non-IID has minimal impact likely generalizes to other multi-class classification tasks where classes have shared features (e.g., network traffic anomaly detection broadly). However, the specific performance numbers are dataset-dependent and should not be directly transferred to production environments without validation.

7 Conclusions

This thesis evaluated Federated Learning for intrusion detection in IIoT environments using the CIC IIoT 2025 dataset (Firouzi et al., 2025), addressing the challenge of building collaborative threat detection systems without centralizing sensitive operational data. The research systematically evaluated FedAvg across 48 configurations covering data heterogeneity, communication parameters, and participation rates (Section 5, Table 2).

The centralized baseline established an upper bound of 0.885 macro F1 (O1). FedAvg under IID conditions closed 48–62% of the centralized–local gap (O2; Figure 1). The key finding is that moderate non-IID data (skew=0.7) degrades performance by only 1.8%, demonstrating FedAvg’s resilience under realistic IIoT heterogeneity (O3; Figures 4 and 5). Convergence analysis revealed that moderate skew primarily lowers the performance ceiling rather than slowing the learning rate, while severe skew affects both. Communication analysis revealed $R = 50$ as the practical sweet spot with diminishing returns beyond (O4; Figure 3), and partial participation at 30% retained 98.6% of full performance (O5; Figure 7).

These results lead to two main conclusions. First, FL-based IDS is viable for IIoT deployments: the performance cost of privacy preservation is manageable (12–38% gap remaining depending on configuration), and the system is robust to the operational constraints (intermittent participation, heterogeneous attack distributions) that characterize real industrial environments. The gap between FL and centralized performance translates to approximately 5–8 additional misclassifications per 100 samples, concentrated in rare attack types rather than high-frequency traffic classes. Second, local computation matters more than communication: increasing local epochs provides more benefit than additional communication rounds, suggesting that investment in edge computing capability is more valuable than network bandwidth for FL-IDS deployments, with $E = 5$ offering the best energy-accuracy tradeoff for power-constrained devices.

8 Future Work and Lessons Learnt

8.1 Future Work

Several extensions could enhance this research:

Advanced FL algorithms. FedProx (Li et al., 2020) adds a proximal term that limits divergence between local and global models, which directly addresses the local model drift observed under severe non-IID conditions (Figure 5). The sharper question is whether FedProx can recover more of the centralized–local gap at severe skew (0.9), where FedAvg currently closes only 28.8%. A focused comparison at that single condition ($K=5$, $E=5$, $R=50$, 3 seeds) would isolate the proximal term’s effect on drift. Adaptive aggregation methods (Reddi et al., 2020) could further improve convergence speed.

Differential privacy. Adding calibrated noise to model updates (e.g., DP-SGD) would provide formal privacy guarantees. Investigating the privacy–utility Pareto frontier across noise scales $\epsilon \in \{1, 5, 10, \infty\}$ would quantify the cost of mathematical privacy on detection performance.

Personalized FL. Fine-tuning the global model on each client’s local data after aggregation (Fallah et al., 2020; Tan et al., 2022) could improve per-site detection, particularly for minority classes that suffer disproportionately under FL (Figure 6).

Device-based partitioning. Using the dataset’s device metadata to partition by device type (sensors, cameras, smart plugs, network equipment) would create a more ecologically valid non-IID scenario than label skew alone.

Larger-scale evaluation. Testing on the full CIC IIoT 2025 dataset (178K+ samples) with more clients ($K \in \{20, 50\}$) would assess scalability to production-sized deployments.

8.2 Lessons Learnt

Local computation is the primary bottleneck, not communication. Prior FL literature emphasizes communication efficiency (compression, quantization), but our results show that insufficient local training ($E = 1$) is a more fundamental failure mode than limited communication rounds. For lightweight IIoT models, the per-round communication cost (~ 44 KB) is negligible compared to modern network capacities; the constraint is whether clients perform enough local computation to produce meaningful updates. The energy-accuracy analysis suggests $E = 5$ as the practical sweet spot for power-constrained devices, capturing 77% of the total improvement at half the computational cost of $E = 10$.

Non-IID impact is domain-dependent. The moderate non-IID penalty observed (1.8% at skew=0.7) contradicts the common assumption that non-IID data severely degrades FL performance. In intrusion detection, attack classes share substantial feature overlap (e.g., DDoS and DoS produce similar network traffic patterns), which may provide implicit regularization that other domains (e.g., medical imaging with distinct pathologies) do not enjoy. The convergence analysis (Figure 5) further clarifies this: moderate skew lowers the ceiling rather than slowing learning, suggesting that the shared feature structure allows FedAvg to converge efficiently even when label distributions differ. This suggests that FL feasibility assessments should be domain-specific rather than relying on generic non-IID benchmarks.

Minority class detection is FL’s weakest point. The per-class analysis (Figure 6) reveals that FL’s aggregate F1 deficit masks a more concentrated problem: already-rare attack types lose the most ground when data is distributed. This has direct implications for future work on personalized FL and client-level oversampling strategies.

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A Appendix: Additional Results

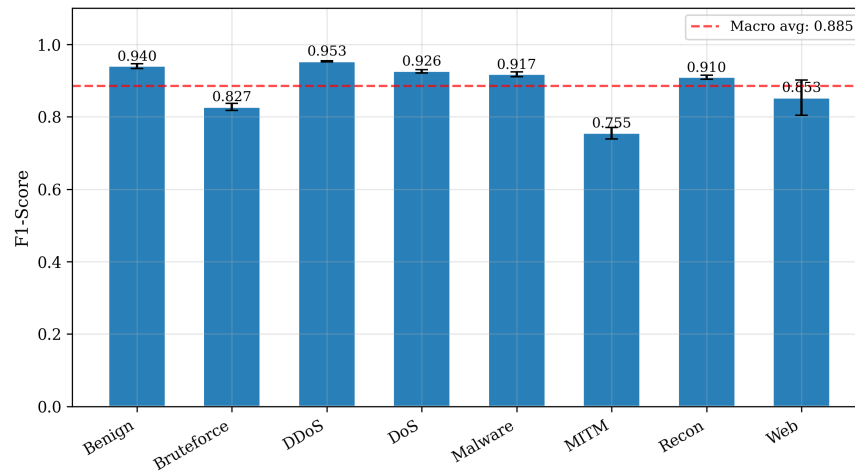


Figure 8 – Per-class F1 for the centralized baseline. Bruteforce and MITM are the hardest classes due to extreme class imbalance (73 and 294 test samples respectively vs. 2,736 for Benign).

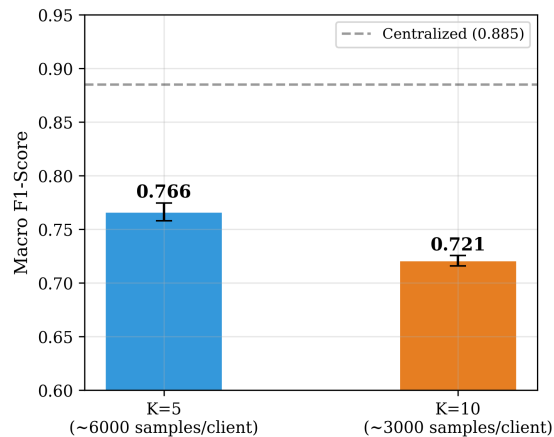


Figure 9 – Effect of number of clients (K). $K = 10$ drops 0.045 F1 compared to $K = 5$, as each client has less data ($\sim 3K$ vs. $\sim 6K$ samples).

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